Procure-to-Pay Internal Audit Report

Report date 17 October 2026 | Purchase-order process, 251,734 items, 76,349 purchase documents

Objective	Test whether the key controls over requisition, ordering, goods receipt, invoicing and payment operated, and report root causes and management actions.
Data	BPI Challenge 2019 event log (4TU.ResearchData, CC BY 4.0): real purchase-order handling at a multinational, 1,595,923 events dated 2018 to 2019 (320 carry other dates in the source), 607 named users. Names, vendors and values are anonymised in the source.
Approach	Risk and control matrix built from the process; ten risks mapped to key controls; six SQL tests run over the full population (not a sample); results rated against the key in section 5.
Overall conclusion	**Needs improvement.** 5 findings need action; 1 rated High.

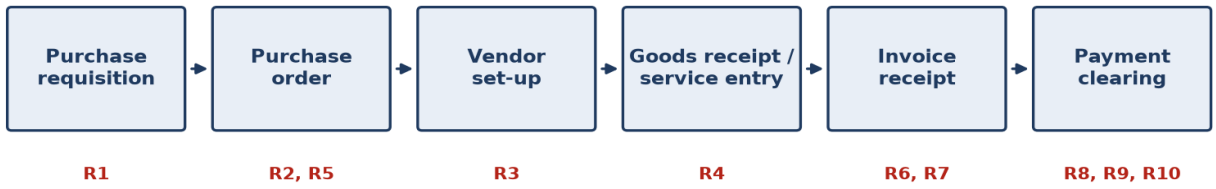
1. Summary of findings

ID	Finding	Rating	Exceptions	Rate	Item value
F1	Possible duplicate payments	High	2,175	3.1%	26,543,453
F2	Buyers recording goods receipts on their own orders	Medium	5,149	2.4%	84,886,857
F3	Orders that may have been split below the approval limit	Medium	5,511	2.6%	34,968,450
F4	Payments made without, or before, a goods receipt	Medium	196	see finding	1,597,698
F5	Large payment batches run by one user	Medium	n/a	86% by one user	n/a

Item value is the purchase-order item value in the log's anonymised units. It shows the amount at stake, not a confirmed loss.

Controls tested with no exceptions: R5 same person orders and processes the invoice or payment (0 of 217,772); R7 goods-receipt-based invoice verification bypassed (0 of 11,128).

2. Process reviewed



Red labels are the risk IDs in the matrix. Vendor set-up sits outside the log, so R3 is documented but not tested.

3. Risk and control matrix

ID	Risk	Key control	Inherent	Test	Rate	Conclusion	Residual
R1	Purchases raised without an approved requisition	Approved purchase requisition required before a purchase order is raised	Medium	Coverage	n/a	Not tested	Medium
R2	Orders split across documents to stay under the approval limit	Approval limits enforced by order value; split-order report reviewed monthly	Medium	T5	2.6%	Needs improvement	Medium
R3	Fictitious or duplicate vendor set up and paid	Vendor master changes approved by a second person; duplicate vendor check	Medium	n/a	n/a	Not tested	Medium
R4	Same person orders and receives the goods	System roles keep purchase order creation and goods receipt with different people	Medium	T3a	2.4%	Needs improvement	Medium
R5	Same person orders and processes the invoice or payment	System roles keep purchase order creation apart from invoice receipt and payment clearing	Medium	T3b	0.0%	Effective	Low
R6	Invoice paid for goods or services never received	System blocks payment until a goods receipt is posted (three-way match)	High	T1a	0.06 %	Effective	Medium
R7	Goods-receipt-based invoice verification bypassed	Invoice receipt blocked until goods receipt on items set up for GR-based verification	Medium	T1b	0.0%	Effective	Low
R8	Supplier paid before the goods are received	Payment released only after the goods receipt date	Medium	T4	0.05 %	Effective	Low
R9	Same supplier paid twice for the same order	Duplicate check on vendor and value before payment; reviewed before each run	High	T2	3.1%	Needs improvement	High
R10	One person runs very large payment batches with no independent review	Payment proposal approved by a second person before release	Medium	Obs	n/a	Not tested	Medium

The matrix with owners, frequency, assertions and test procedures is in P2P_Risk_Control_Matrix.xlsx. R1: a requisition event exists for 46,592 of 251,734 items, so the requisition control cannot be tested from this log. R3: vendor master changes are not in the log.

4. Detailed findings

F1 Possible duplicate payments (Rating: High; risk R9)

Observation	2,175 of 70,817 paid items (3.1%, item value 26,543,453) share vendor, company and value with an item in a different purchase document created within 7 days, and both were paid. Values paid five or more times by one vendor are treated as recurring prices and excluded. The 238 vendors involved are concentrated: the top five hold 57% of the value.
Expected	Each supplier invoice is paid once. A duplicate check on vendor and value runs before every payment run.
Probable root cause	Probable: the system checks duplicates on invoice number only, so the same order keyed twice under different document numbers passes. Confirm with Accounts Payable.
Risk	Cash paid twice for one purchase; recovery depends on supplier goodwill.
Recommendation	Add vendor, company and value to the pre-payment duplicate check. Review the listed items, recover any confirmed duplicates, and block repeat entries within 7 days without a second approver.
Proposed management action	Run the duplicate review on the listed items and report recoveries; configure the extended check in the payment proposal.
Action owner and due	Accounts Payable Manager | due 01 Dec 2026

F2 Buyers recording goods receipts on their own orders (Rating: Medium; risk R4)

Observation	5,149 of 217,772 items (2.4%, item value 84,886,857) were created and goods-received by the same named user. 47 users are involved; user_000 alone accounts for 26% of the items. No item had the same person create the order and record or clear its invoice.
Expected	Creating a purchase order and confirming receipt of the goods are done by different people.
Probable root cause	Probable: purchasing and receiving roles are combined for some users, possibly in small sites or for services where the buyer confirms delivery. Confirm with IT Security.
Risk	A buyer can order, receive and so trigger payment for goods that never arrived.
Recommendation	Remove the combined access or, where the team is too small to separate, add a monthly review by a manager of every item the buyer both ordered and received. Start with the top users listed in the workbook.
Proposed management action	Run an access review of the listed users; split roles or document a compensating monthly review.
Action owner and due	IT Security Manager | due 16 Dec 2026

F3 Orders that may have been split below the approval limit (Rating: Medium; risk R2)

Observation	5,511 items (2.6%, item value 34,968,450) belong to groups where one buyer raised two or more documents to the same vendor on the same day, each between 4,000 and 10,000 and together at or above 10,000. The approval limit is an assumption because the log holds no approval matrix.
Expected	Order value, not document count, decides who must approve.
Probable root cause	Probable: approval is checked per document, so several smaller documents avoid the higher approval level.
Risk	Spend above a buyer's limit is committed without the approval the policy requires.
Recommendation	Aggregate same-vendor, same-day documents when applying approval limits and add a monthly split-order report reviewed by the Procurement Manager. Test the report against the actual approval matrix.
Proposed management action	Build the split-order report and review the listed groups against the approval matrix.
Action owner and due	Procurement Manager | due 16 Dec 2026

F4 Payments made without, or before, a goods receipt (Rating: Medium; risk R6, R8, R7)

Observation	109 items (0.06%, item value 470,856) had an invoice cleared with no goods receipt or service entry sheet ever recorded. 49% of these had their goods receipt cancelled after the invoice was posted, and the payment was still released. A further 87 items (0.05%, item value 1,126,842) were cleared before their first goods receipt. On the 11,128 items set up for goods-receipt-based verification, 0 had an invoice posted ahead of the receipt.
Expected	An invoice is paid only after the goods or services are received and the invoice matches the order and the receipt.
Probable root cause	Probable: manual unblocking of invoices held for goods receipt, or receipts recorded outside the system. Confirm with Accounts Payable.
Risk	Payment for goods or services not received.
Recommendation	Review each listed item for evidence of delivery. Restrict manual unblocking of payment blocks to a named person and report every unblocked invoice weekly.
Proposed management action	Obtain proof of delivery for the listed items; restrict and report manual unblocking.
Action owner and due	Accounts Payable Manager | due 16 Nov 2026

F5 Large payment batches run by one user (Rating: Medium; risk R10)

Observation	user_002 cleared 166,353 of 194,393 invoice-clearing events (86%), with up to 4,038 in a single minute. This shows a payment run executed under one named user. The log does not show whether a second person approved each run.
Expected	A payment proposal is approved by someone other than the person who runs it.
Probable root cause	Probable: payment runs are scheduled under a named user's login instead of a controlled batch account. Confirm with the Finance Controller.
Risk	One credential can release all supplier payments in a run.
Recommendation	Run payments from a batch account, keep the proposal approval as a system record, and review the approver against the operator each month.
Proposed management action	Move runs to a batch account and evidence the second approval.
Action owner and due	Finance Controller | due 01 Dec 2026

5. Rating key, scope and limits

Measure	Rule
Inherent rating	Likelihood x impact (each 1 to 5): High 15 or more, Medium 8 to 14, Low below 8. Likelihood and impact are reviewer judgement.
Control conclusion	Exception rate up to 1% Effective, up to 5% Needs improvement, above 5% Ineffective. No test: Not tested.
Residual and finding rating	Effective control lowers the inherent rating one level; otherwise unchanged. A finding takes the highest residual rating of its risks.
Overall conclusion	Unsatisfactory if two or more findings are High; Needs improvement if any finding is rated; otherwise Satisfactory.

Limits. The log holds one item value per event, not separate goods receipt and invoice amounts, so three-way match is tested on event sequence and value at risk is the item value. The approval limit in the split-order test and the 7-day window in the duplicate test are reviewer assumptions. Invoice numbers are not in the log, so duplicate hits are leads to verify with source documents. Root causes are inferred from the pattern and must be confirmed with the process owner, and management actions are proposals until accepted.